8.1.1 The multiplication of two square matrices.

A] Algorithm

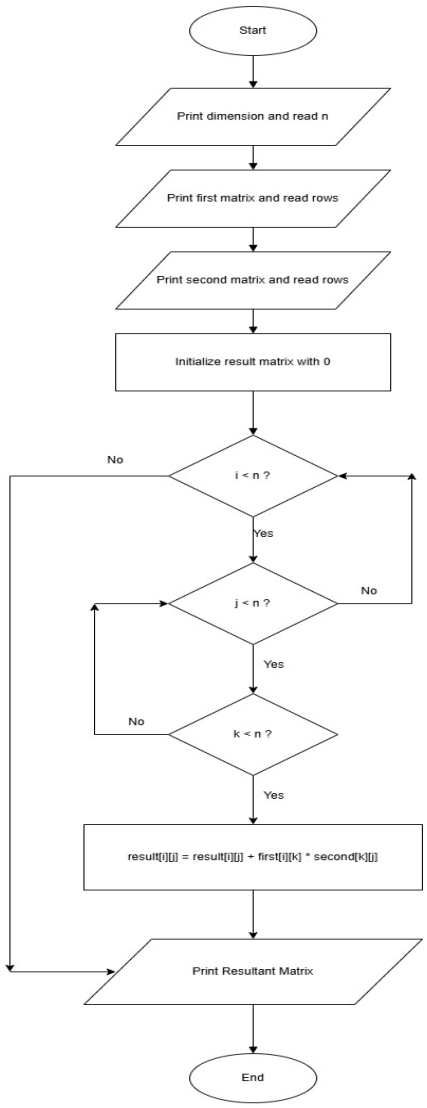
Step 1: Start
Step 2: Input the dimension a of the square matrix.
Step 3: Display "first matrix".
Step 4: Initialize an empty list A.
Step 5: Repeat a times to read rows of the first matrix and store them in A.
Step 6: Display "second matrix".
Step 7: Initialize an empty list B.
Step 8: Repeat a times to read rows of the second matrix and store them in B.
Step 9: Create an empty matrix result of size a × a and initialize all elements to 0.
Step 10: For i from 0 to a-1
Step 11: For j from 0 to a-1
Step 12: For k from 0 to a-1
Step 13: Calculate
result[i][j] = result[i][j] + A[i][k] * B[k][j]
Step 14: Display "Resultant Matrix".
Step 15: Print each row of the result matrix.
Step 16: Stop.

B] code

```
a=int(input("dimension: "))
print("first matrix:")
A= []
for i in range(a):
        r=(list(map(int,input().split())))
        A.append(r)
print("second matrix:")
B = []
for i in range(a):
        r = list(map(int,input().split()))
        B.append(r)
result=[]
for i in range(a):
        result.append([0]*a)
for i in range(a):
        for j in range(a):
                for k in range(a):
                        result[i][j]+=A[i][k] * B[k][j]
```

```
print("Resultant Matrix:")
for row in result:
        print(*row)
```

C] flowchart



D] output

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8.1.1. Multiplication of Two Square Matrices

Write a Python program to perform the multiplication of two square matrices. The program should take two matrices as input and output their product.

Input Format:

- The first line prints the "dimension: " and reads an integer n, representing the dimensions of the square matrices $(n \times n)$.
- The second line prints: "first matrix:" and then reads n lines, each containing n integers separated by spaces, representing the rows of the first matrix.
- Next line prints: "second matrix:" and then reads n lines, each containing n integers separated by spaces, representing the rows of the second matrix.

Sample Test Cases +

Multiplica... Submit

```python
a=int(input("dimension: "))
print("first matrix:")
A= []
for i in range(a):
    r=(list(map(int,input().split())))
    A.append(r)
print("second matrix:")
B = []
for i in range(a):
    r = list(map(int,input().split()))
    B.append(r)
result=[]
for i in range(a):
    result.append([0]*a)
for i in range(a):
    for j in range(a):
        for k in range(a):
```

Terminal Test cases

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